15-6

Vertebral Column

→ Consist of 33 Vertebrae and 23 intervertebral discs.

→ Vertebrae Column, divided into 5 regions

Cervical - 7
Thoracic - 12
Lumbar - 5
Sacral - 5 (fused)
Coccygeal - 4

→ Vertebras ↑s in size from Cervical to lumbar region and ↓s in size from Sacral to Coccygeal region

→ Fetus - One long curve, Convex posteriorly

→ Infants - Secondary Curve.

→ Adult - 4 distinct antero-posterior curve.

2 curves (thoracic & Sacral) - posterior convexity are primary curves.

2 Curves (Cervical & lumbar) - posterior concavity are Secondary curves.

→ Posterior convexity / Anterior Concavity - Kyphotic Curve.

→ Posterior Concavity / Anterior Convexity - Lordotic Curve.

Mobile Segment :-

→ Smallest functional unit in the Spine is the mobile Segment.

Typical Vertebra :-

→ Structure - 2 main parts.
— Anterior, cylindrically shaped vertebral body
. Posterior, Irregularly shaped vertebral or Neural arch.

→ Vertebral body - wt bearing column for Spine.

→ Neural arch

- Pedicles
- Posterior element

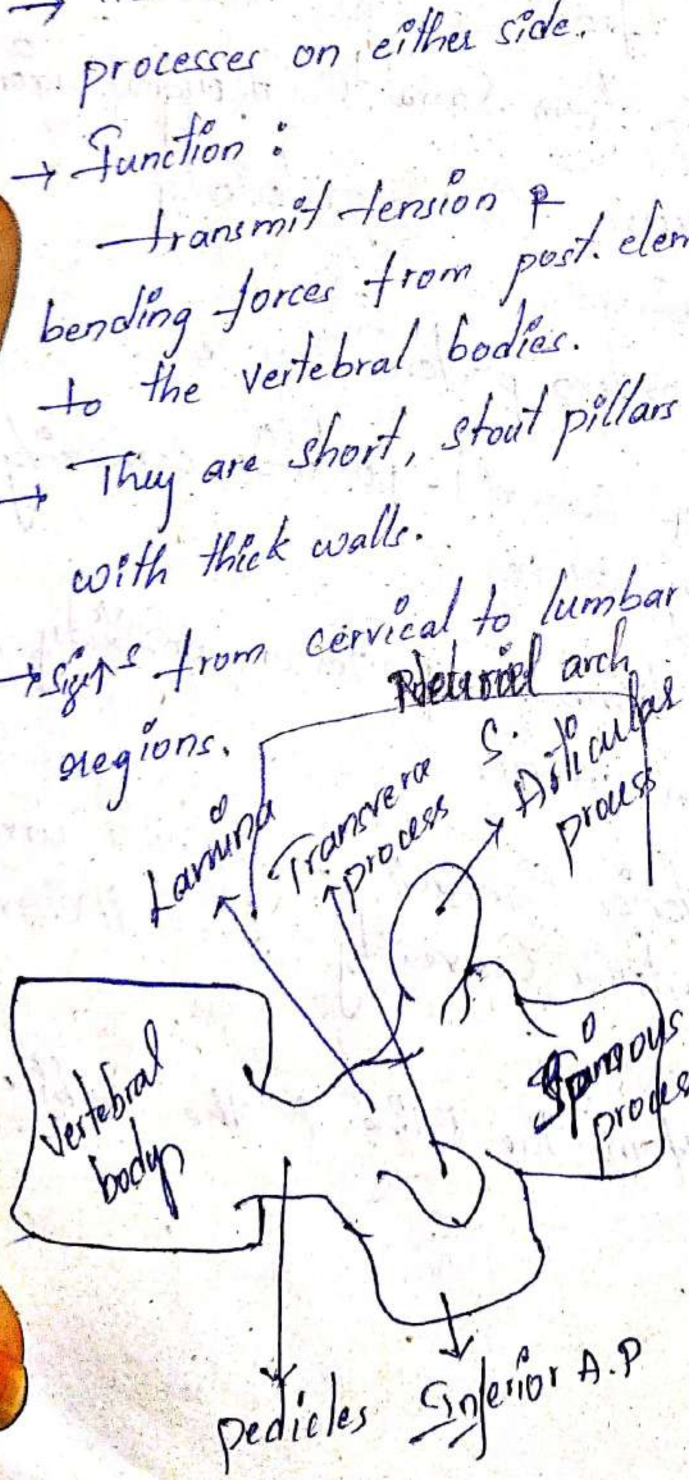
Pedicles

→ lies anterior to the articular processes on either side.

→ Function :
— transmit tension & bending forces from post. element to the vertebral bodies.

→ They are short, stout pillars with thick walls.

→ Size↑s from cervical to lumbar regions.



Posterior element

→ Laminae, Articular processes, Spinous process, transverse processes.

→ Laminae :
Centrally placed.

→ Laminae are thin, Vertically oriented piece of bone that serve as "roof" to Neutral arch, protects the spinal cord

→ Laminae transmit forces from post. element to the pedicle, through them to verterbral body.

→ Force transfer occurs through a region of the laminae called Pars Interarticularis

→ Present b/w the laminae and articulae processes.

→ Pars Interarticularis, that transmitted
- bending forces ~~transfer occurs~~
from the vertically oriented lamina to
horizontally oriented pedicles.
- They are more developed in lumbar
region.

→ Spinous processes & 2 transverse
processes are sites for muscle attachment
→ Articular processes consist of 2 superior
and 2 Inferior facets for articulation
with facets from Cranial & Caudal vertebra

Inter vertebral Disc -

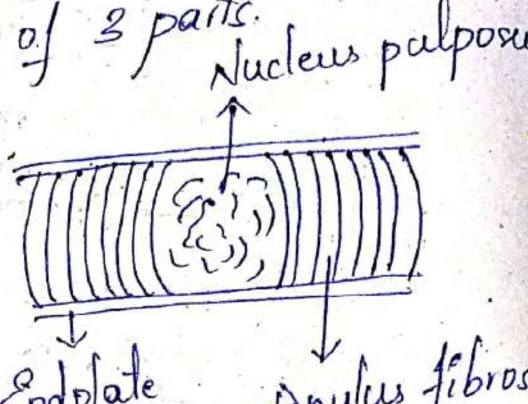
→ Has 2 principle functions
to seperate 2 vertebral bodies, thereby ↑ing available
motion & to transmit load from one vertebral
body to next.
→ Make up about 20 to 33% of length of Vertebral
Column, ↑s in size from cervical to lumbar region.
→ Disc thickness - Cervical : 3 mm (wt bearing are lowest)
Lumbar : 9 mm (wt bearing are highest)
→ Disc are smallest in cervical & greater/largest
in lumbar region.
Ratio b/w disc thickness and vertebral body height.
→ Greater the ratio and Greater the mobility.

→ Ratio is greater in cervical and lumbar, which produces greater mobility.

→ Ratio is Smallest in thoracic region, which produces with greater stability.

→ Intervertebral disc composed of 3 parts.

1) Nucleus pulposus
2) Anulus fibrosus
3) Vertebral End plate



→ Nucleus pulposus - Gelatinous mass at center.

Anulus fibrosus - fibrous Outer ring.

Vertebral Endplate - Cartilaginous layer covering the Superior and inferior Surface of disc.

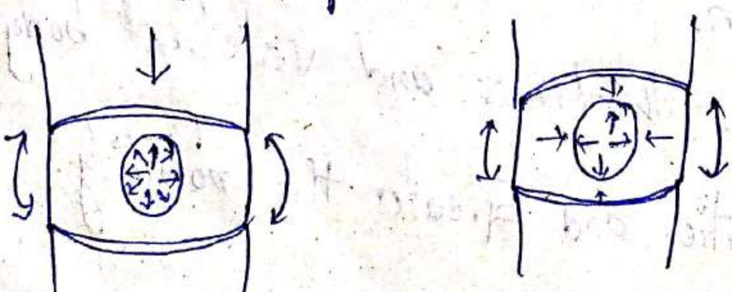
Nucleus Pulposus -

→ It consist of 70 - 90% of water by Age & time.

→ 65% of Proteoglycans (PGs) - Holds water

→ 15-20% of Collagen, Proteolytic enzymes, Chondrocytes elastin fiber, proteins.

→ Collagen → Type I → ↓ (distraction compression)
Type II → ↑ (Retraction compression)



Anulus Fibrosus -

→ Water (60 - 70%), Collagen (50 - 60%), ↑ PGs proteoglycans 20% dry wt, Elastin - 10% fibroblasts Chondroblasts.

Collagen

Type I ↑↑ | Type II

- Resists tensile forces

→ Superior and Inferiorly End plate.

End Plate -

→ Layer of cartilage superior / Inferior surface (0.6 - 1 mm) thick.

→ It has PGs, Water, Collagen.

→ Cartilage Cells → Hyaline, → Fibrocartilage

Hyaline $\xrightarrow{\text{Age}}$ Fibrocartilage

Hyaline ↓ Near Vertebra

Fibrocartilage ↓ Near Nucleus.

→ Fibrocartilage helps to take up load.

Innervation - Intervertebral disc are innervated in the Outer 1/3 - 1/2 of Anulus fibrosus branches of vertebral and Sinuvertebral Nerves.

↳ Also Innervates some Ligaments.

Nutrition -

Metaphyseal arteries from dense capillary plexus in base of end plate cartilage and Sub-chondral bone

- Also by diffusion.

Articulations -

→ Two types of joint articulation

1) Facet jt

2) Inter body jt

Facet jt

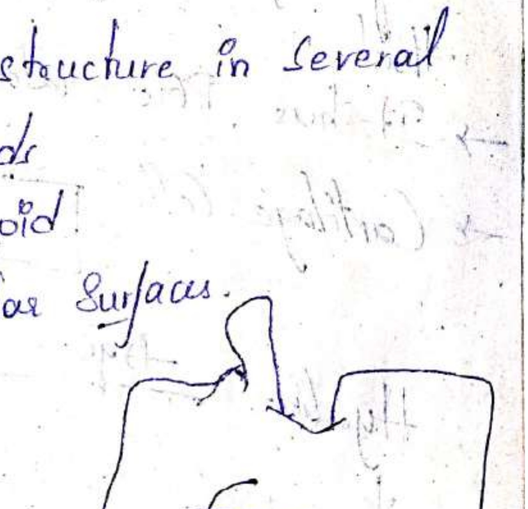
Also known as Zygopophyseal Articulation and Diarthrodial joint.

→ Intra-Articular accessory structure in several types

→ Adipose tissue pads

→ Fibroadipose Miniscoid

→ Involved in protecting articular surfaces.



Inter body jt

→ Side to side translation.

→ Superior Inferior translation

→ Anterior + posterior translation

→ Side to side Rotation

→ Superior ~~to Posterior~~ Inferior (Rotation in transverse plane)

→ Anterior posterior Rotation.

Ligaments

Anterior Longitudinal Ligament :-

→ Starts from Sacrum to C_2 [C_2 - Occiput]

Called as Anterior Atlanto-Occipital.
Anterior Atlanto - Axial

→ 2 Layers

Superficial - Long, cover several vertebra.

Deep - Short, - covers single vertebra.
- Blends with Anulus fibrosus
- Reinforces Antero-Lateral portion of Disc/Body.

Well developed in [cervical, Lumbar

→ ↑↑ tensile strength at cervical - 2x than posterior Longitudinal Ligament.

→ Compressed in flexion

Stretched in Extension

Lax in Neutral.

Posterior Longitudinal Ligament :-

→ From Sacrum to C_2 [C_2-Occiput]

Called as Tectorial membrane.

→ 2 Layers

Superficial - Long, Covers many vertebra

Deep - Fibers extend to adjacent vertebra.
- Outer layer of Anulus and end plate margins.

→ Lumbar region ligament is narrow - No proper support for disc → High incidance of Herniation in posterio-lateral direction.

Ligamentum Flavum :-

→ Lamina ↔ Lamina (Thick, elastic).

C_2 - Sacrum.

→ Smooth posterior surface of canal.

C_2 to Occiput - posterior Atlanto occipital

- Atlanto axial membrane

→ Creates compressive force on disc

Stiff disc - Better support will not buckle on itself [Elastic]

Inter Transverse Ligament :-

→ Extremely variable.

→ Pass b/w transverse process and attach to deep muscles [Stretch - compress] - with Lateral flexion.

Interspinous Ligament :-

→ Connects the spinous process of Adjacent vertebrae.

→ 1st one to damage with excessive flexion.

Contribute to lumbar spine stability and degenerative aging [Collagen type I, PGs, Elastin].

Supraspinous Ligament :-

→ Strong, Cordlike, Connects spinous process
[$C_7 - L_{3-4}$]

→ Merge with thoraco-lumbar fascia.
Cervical region - Ligamentum Nuchae

→ 1st one to damage with excessive flexion
They have mechanoreceptors that affect
Multifidus → Recruitment of Spinal stabilizers.

Kinematics :-

→ Movement occur in interbody jt and zygophyseal jt.

→ Movements → Flexion
Extension
Lateral flexion
Rotation

→ Coupling movements occurs in Lateral flexion + Rotation.

→ What affects the motion orientation of Articulating facet.
Elasticity, Fluidity, Thickness of disc.
Extensibility of Muscle
Extensibility of Capsule
Extensibility of Ligament

Facet
→ Direction decided by orientation of facet.
changes in every region.

Saggital plane - Flexion, Extension [Lumbar region]
Frontal plane - Lateral flexion [Thoracic region]
Transverse plane - Rotation [cervical region].

Disc

→ Amount of motion determined by size.
Nucleus acts like a pivot with lot of fluid.

Flexion

→ Anterior tilt + Glide of the spine [Anterior glide]
→ Flexion & extension is more pre-dominent in lumbar region with saggital plane.
→ Size ↑ in intervertebral foramen.
→ Anterior anulus Compressed.
→ Resistance for flexion by spinous ~~pr~~ ligament, Capsule, ligamentum flavum, posterior anulus, posterior longitudinal ligament.

Extension

→ Posterior tilt + posterior glide
→ Size ↓ in intervertebral foramen
→ Posterior anulus Compressed.
→ Resistance - Spinous process.
- Anterior trunk muscles, Anterior Longitudinal Ligament → Only ligament that resists

extension, thats y thick and strong.

Rotation

→ Depends highly on facet and varies from region to region.

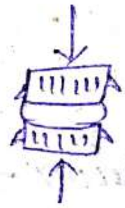
Lateral flexion

→ Lateral tilt + Rotation + Translation

→ Inter vertebral foramen ↑s in opposite side
↓s in same side.

→ Anulus compresses on the same side and stretches on opposite side.

Kinetics :-



Axial Compression

→ Gravity, GRF, Muscles and Ligaments

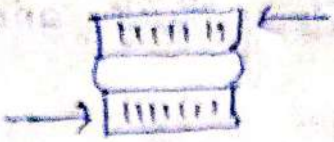
→ In Vertebral body [25-55%] load is taken up by Cancellous bone before 40 yrs, After 40 yres cortical bone takes higher loads.

→ Zygophyseal take [0-33%] depending on posture & region.

→ Nucleus pulposus exhibits swelling pressure outward countered by anulus and end plate [Equilibrium].

→ Disc exhibits creep with loading and diurnal changes.

Shear force



→ Acts on midplane of disc.
→ Vertebrae undergo translation.
→ L-Spine facet resists Shear and then the disc.

Torsion

→ Axial rotation [Coupled motion]
→ Tortional stiffness determined by facet, Cortical bone, disc - Anulus resists according to orientation in Lumbar region.
→ ↑ Risk of Rupture + flexion + Compression.

Bending

→ Compression + Tension force
↑ Anterior Structures ↑ Posterior Structure
↳ For flexion, opposite for extension.

→ Creep exhibited by Structure.
If it goes past Toe region of Stress - Strain curve Structure wont return to orignal dimention
[↑ Risk of injury]

→ Lateral Bending - Contralateral Inter transverse Ligament provides Stability.

Cervical Spine :-

→ divided into 2 region.

Upper Cervical Region [Craniovertebral]

Lower Cervical Region.

→ They are 2 types of Vertebrae

Typical Vertebrae

Atypical Vertebrae

C_1 - Atlas

C_2 - Axis

C_7 - Transition vertebra where the cervical spine turns into thoracic spine.

Atypical Vertebra

Atlas - C_1

→ Function - Cradles occiput and transfers forces to the lower cervical vertebra.

→ No body ; 2 large facets

Foramen for Vertebral artery

Superior Articulating Surface - Large , Concave

Inferior Articulating Surface - Slightly Convex

Facet on internal surface → for Dens.

Axis - C_2

→ Transmit force, provides axial rotation dens has

Superior facet which articulates with anterior arch of Atlas Superior facet faces upward, Laterally
Inferior facet faces Anteriorly.
→ Spinous process has a bifid (2 part)

Articulation

Atlanto-Occipital - (2) jts

→ Concave Superior facet (Atlas) and Convex Condyles of occiput.
→ True Synovial jt, Almost in horizontal plane.
→ It have intararticular menicoid and fibro adipose structure.

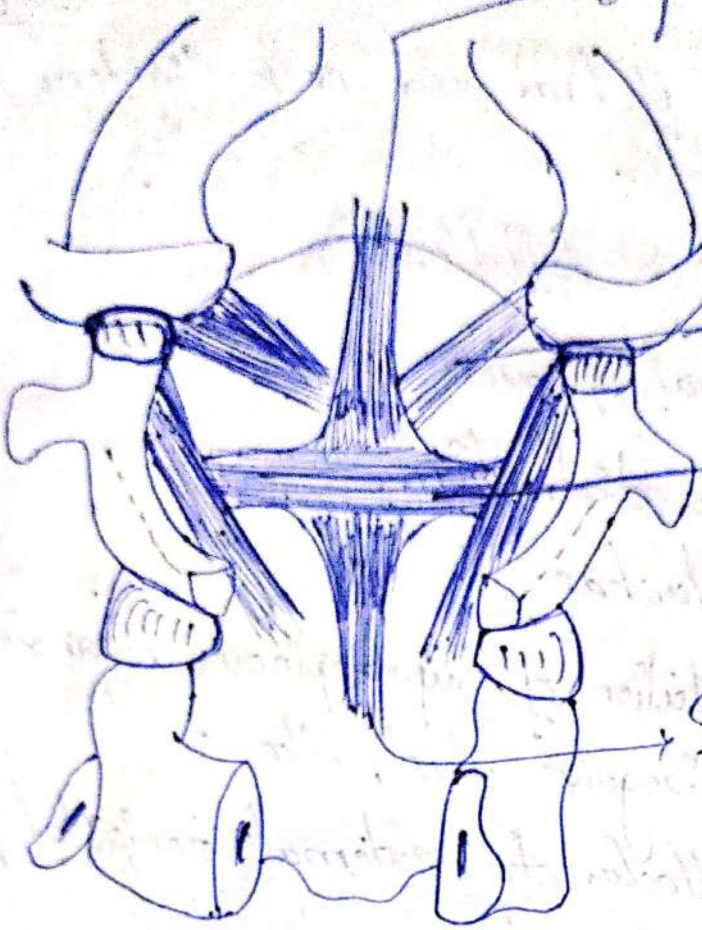
Atlanto-Axial Jt (3)

→ Median Atlanto axial jt [Synovial trochoid (pivot)]
B/w Dens and Atlas.
→ Lateral jts b/w Superior facet of axis and inferior facet of atlas - Biconvex
→ Axis rotates B/w osteoligamentous ring formed by anterior arch and transverse ligament.

Ligaments

Transverse Atlantal Ligament

→ Across ring of atlas → posterior (Spinal cord)
↳ Anterior (dens)



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→ Anterior Surface has cartilate
Superior long ligament → Occiput.
Inferior fibers → posterior to Axis

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Atlantal Cruciform Ligament

→ Anarticular Surface holds Dens prevents anterior displacement of C_1 on C_2 [Stability]
→ Superior and inferior parts assist
Very strong - dens break before Ligament

Alar Ligament

→ Paired Ligament from either side of dens → Occiput Condyles
↓
Atlas laterally

1cm long, pencil width. Lax - Neutral, Taut - Flexion.

→ Axial rotation of head and neck tightens the alar ligament.

→ Right (upper) Left (lower)

→ Prevent lateral flexion.

→ Prevents $C_1 - C_2$ distraction.

<u>Ligamentum Nuchae</u>

→ It is the evolution of Supraspinous ligament.

→ It is thick, triangular, sheet, like

Superiorly it attaches to external occipital protuberance and foramen magnum.

Inferiorly - Tip of C_7 spinous process.

→ They are 2 parts
  Dorsal Raphae
  Ventral

→ Dorsal Raphae - Narrow strips of collagnous tissue interlacing tendons of upper traps splenius capitis and Rhombod minor muscle.

→ Ventral midline Septum Runs ventrally on doral raphae, Continuous with inter spinous lig (A-A) membrane, fascia of Semi spinalis capitis, Splenius capitis.

→ Attached to vertebra and is capable of R flexion.

Longitudinal Ligament

→ Ligamentum flavum - posterior Atlanto-occipital and atlanto-axial membrane.

→ Less elastic, ↑ ROM (Rotation)

→ Anterior Atlanto-Occipital and Atlanto-Axial membrane.

→ Tectorial membrane continues as posterior longitudinal ligament.

↳ Broad strong membrane originates from posterior vertebral body of axis, covers dens & Crutiate ligament and inserts at

Lower Cervical Region

Typical Vertebra - Body

→ It is small and transverse diameter is high than ~~posteri~~ Anterio-posterior diameter.

→ Uncinate process -

After the body there is the upward curve which makes the concave shape in the frontal plane is made by this uncinate process.

Arches

Pedicles - After the body posterio-laterally.

Lamina - Posteriorly but medially.

→ They are thin and slightly curved.

Facit

Superior facit - Oval shaped and superoposterior.

Inferior facit = Inferioarly anterior.

→ Facit size ↑s as you go down from $C_3 \rightarrow C_7$

Transverse Process

Foramen - Vertebral artery, Vein, Venous plexus passes.

Spinous Process

→ It is short, ~~Sp~~ slender, bifid tip.

Disc

Anular fibers - Cresent shape

- It is thick anteriorly
  Taper laterally
  posteriorly only with posterior longitudinal ligament

Joints of Luschka

- Also known as Uncovertebral joints.

→ Fissures in disc develop along with the Uncinate process.

↓

- These clefts becomes joint.

Kinematics

Upper Cervical Spine

Atlanto-Occipital joint

→ Nodding movement.

Flexion - Condyles rolls forward and slide backward.

Extension - Condyles rolls backward and slide forward.

→ Zero degree of rotation and lateral flexion, they are resisted as Condyles rise up from Atlantal Socket.

Atlanto-Axial joint

→ All movement ↓ but main movement rotation ↑.

→ Atlas pivots around dens to 45° both sides.

→ Alar ligament limits the motion (Resistated).

→ Lateral flexion coupled with contralateral rotation.

Lower Cervical Spine

→ Movement depends on shape and orientation of the facet.

→ Flexion - Anterior tilt + Anterior translation.

→ Extension - Posterior tilt + Posterior translation.
→ Lateral flexion Coupled with ipsilateral rotation.
→ ROM :- Flexion - Extension, ↑ C_2/C_3 → C_5/C_6

Joint capsule is lax (↑ ROM in young)

Disc Height > Disc width → ↑ ROM

Disc (C_5 - C_6) → ↑ Stress

Kinetics

→ Cervical region bears less weight but it is more ~~sto~~ mobile.

Upper Cervical Spine (No disc).

→ Forces transferred through facet, lamina and arches.

→ Lamina of axis and C_7 are heavily loaded with trabeculae → absorption of force.

Lower Cervical Spine

3 - Columns - 1 Anterio central
- 2/3 force (vertebral body + Disc)

- 2 Posterio laterally
- 1/2 force (facet)

→ Compressive force ↑↑ (flexion + extension)

→ Extension - Resisted by facet block and inter vertebral disc.

→ Flexion - Resisted by posterior longitudinal ligament and spinous ligaments.

Muscles

Posterior - Trapezius, Levator Scapulae, deep muscle

Lateral - Scalenes, Sternocleidomastoid

Anterior - Rectus Group.

Thoracic Spine

Articulation

Interbody joint

→ Flat vertebral surface allows all translation disc between allows tipping

→ Cervical region
Saddle type joint Concave + Convex
↓
Due to uncinate process.

Facet joint

→ Plane synovial jts.
Fibro adipose miniscoid tissue

→ ↑ Lateral flexion, Rotation ↓ Flex - Extension
Capsule around it more taut than cervical.

Ligaments

- Ligamentum flavum and anterior longitudinal Ligaments are thicker in thoracic Region.

Body

- Anterio-posterior = Transverse diameter [= Equal]
  ↓
  Gives Stability.
- Wedge shaped → produces kyphosis
- Costocapitular demifacet.

Facet

- Frontal plane orientation.
  Transition at T_{10} - T_{11}
- Superior facet - posteriorly, Superiorly, Laterally
  Inferior facet - Anteriorly, Inferiorly, Medially.
- Transition to saggital plane slowly ~~Superior~~ →
  Superiorly - ~~Anterio to~~ Posterolaterally
  Inferiorly - Anteriomedially.

Transverse Process and Disc

- Thick ends - Costotubercular articulation with ribs [oval facet]
- Disc are thin - ↑ stability ↓ mobility.

→ Wedge-shaped - kyphosis - 1° stabilizer.

~~Disc~~ Spinous Process

→ slope inferior from $T_5 - T_8$

$T_{11} - T_{12}$ - triangular

- Horizontal

Kinematics

→ Upper Thoracic Region $[T_1 - T_6]$

→ Flexion - Extension limited due to rigid rib cage, ligament facet orientation.

→ ↑ Lateral flexion + Rotation → Coupled movement of rotation + lateral flexion - Same side

→ Lateral flexion and rotation limited by rib cage

Lower Thoracic Region $[T_9 - T_{12}]$

→ ↓ Rotation and ↑ flexion - Extension = facet in Saggital plane.

→ Coupled movements of Rotation + lateral flexion (opp side)

→ Ribs on ipsilateral side of rotation ↑ Convexity

Rotation ROM depends on Ribs flexibility, ROM at [Costo vertebral]

Kinetics

→ ↑ Compressive forces

Kyphotic, upper limb

→ Line of gravity - Anterior

↳ Flexion moment (thoracic spine)

Counteracted by posterior longitudinal ligament and spinal extensors.

Lumbar Spine

Body:

→ Trasverse diameter > Anterio posterior diameter > height

↑ size.

Arches:

→ Pedicle are short, thick.

→ Lamina is short & broad.

Foramen: - Triangular large

Spinous process :- Broad, thick, Horizontal.

Facet

Facet vary in both shape and orientation - Saggital

→ Mamillary process - Multifidus and medial inter transverse (attachment)

→ Curved surface, Biplanar
Anterior - frontal plane
Posterior - Saggital plane
↑ flexion - Extension
↓ Rotation.

Disc

→ Lamella - has Concentric rings.
120° of opposite direction.
→ Helps in resist tensile force in all directions.
→ Lagest disc.
Eliptical + Concave (Posterior)
Thoracic are convex (Posterior)

Ligaments

→ Supraspinous Lig - till L_3/L_4
↳ Deep fiber reinforced by Multifidus.
→ Infra - Transverse Lig - Not a true ligament
↳ It replaced by illioLumbar ligament at L_5.
→ Posterior Longitudinal Lig - Thin.
→ Ligamentum flavum - Thick
→ Anterior Longitudinal Lig - Thick, Strong, well developed.

Iliolumbar ligaments

Anterior

→ L_5 transverse process of illiac Crest.

Posterior

→ L_5 transverse process of illiac Crest (posteriorly side)

Sacral Band / lumbosacral ligament

→ Transverse process of L_5 → Ala of Sacrum and anterior sacro-illiac ligament

Stabilizes (L_5)

→ Resist all movements of L_5 to S_1 (anterior displacement)

→ This lig maximally loaded in lumbo-sacral junction in absence of muscle protection

Slouch sitting → Creep → Injury → Lower back pain.

Thoracolumbar Fascia

Anterior (passive)

→ Arise from quadratus lumborum, Joints the middle layer and inserts into inter-transverse ligament.

→ Through transverse process of lumbar Spine.

→ Function:

Transmits tension to Spinous process by Contralateral Hip extension.

Posterior (Active layer)

→ Large, thick, fibrous, Arise from spinous process

→ Travels laterally over erector spinae → Lateral Raphe

Internal oblique, transverse abdominous, arise from Lateral Raphe. ~~arise from~~

→ Goes back to transverse process.

Middle

→ Both the layers are surround the lumbar extensor group.

Uses

→ ↑ Spinal stiffness.

→ Stabilizing Corset → Hoop around abdominus

~~activati~~

→ Transverse abdomines activation → Stiff fascia

Mechanical transmission through (Gluteus and latissimus dorsi) → B/w trunk and pelvis.

Kinematics

Lumbopelvic Rhythm

→ Lumbar flexion + Anterior - pelvic - tilt

① Lumbosacral flexion ⎤ → Bending

② Anterior pelvic tilt at hip jt ⎦ down.

→ Vice - versa while Coming up.
→ Hip motion ↑ ROM, ↓ flexibility required.
→ Prevents excessive lengthening of anulus and posterior ligament [protective function].

Vertebral Motion

Flexion → Anterior tilt + Anterior glide.
Extension → Posterior tilt + Posterior glide.
→ Disc will Compress anteriorly - Nucleus will more posteriorly.
→ In extension vice - versa.
↳ Inter vertebral space ↓
↑ (flexion - extension) at lumbo sacral region
↑ lateral flexion - Rotation - Upper lumbar region and starts reducing in lower lumbar region.

Kinetics

→ Compression - 80%
→ Primary function of lumbar region is to provide Support to upper body (static + Dynamic).
→ Tremendous compressive force ← Muscle contra
→ Compressive / Shear forces can change with Lordosis.

→ Compressive load on facet
Degeneration of Disc

Shear force - 20%

→ Present in upright position.
Anterior shear force - Lordosis
Resist by inferior facet of Superior vertebra
Over Superior facet of inferior vertebra.
→ facet orientation → Big roll.
→ If facet is in saggital plane. The capsule and ligament will be stretched ← Creep.
→ Some compressive forces on facet even if its biplanar → Pain, Erector Spinae provides dynamic restrain to ↑ Anterior Shear.

Sacral Region

Structure

Sacral Articulation

→ Ear shaped, Laterally ($S_1 - S_3$)
→ Surface is multiplanar, Curvy covered with hyaline Cartilage

1-3 mm thickness > Illiac cartilage.

Articulation of Illia

→ Hyline Cartilage (↑ density) (↓ thickness)

→ Surface is Curvy, multiplanar posteriorly and superior tuberosity articulates with Sacral tuberosity via Sacro-illiac interosseous Ligament.

Tuberosity - Irregular, Bony Surface (posterior) to articular Surface [Syndesmosis].

Stability

→ SI jt relives stress on pelvis ring allows forces to move through lumbar region.

Mobility

→ Allow movement to absorb the twisting forces from lower limb.

Sacroillioc Ligaments

→ They are 3 parts

1) Anterior / capsular ligament
2) Interosseous - major SI jt link b/w tuberosities
3) Posterior → strength & stability
   ↳ Long posterior SI ligament.

→ Re-inforced by Quadratus lumborum, Erector Spinae, Illiacus piriformis, Glut. max, minimus

Iliolumbar Ligaments

→ Anterior, posterior, sacral band. Attaches from lumbar vertebra → Illium

Sacrospinous Ligament

→ Ischial spine → Sacrum, Coccyx forms the inferior border of great sciatic Notch.

Sacrotuberous Ligament

→ Ischial tuberosity → Sacrum, Coccyx ; inferior border of lesser sciatic notch.

Symphysis Pubis:

Superior pubic ligament - Very thick, dens, and Supports the superior ascept.

Inferior pubic ligament - Present on the inferior ascept of the pubic and Ramus.

Posterior Pubic Ligament - Articulation reinforced by Rectus abdominus, tranverse Abdominus, adductor long. Oblique.